

BHARATH MYSORE

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STORAGE INFRASTRUCTURE AND DISTRIBUTED SYSTEMS

Systems software engineer and technical leader with 20+ years of experience across cloud storage, operating systems, and developer tools. At Oracle Cloud Infrastructure, design and implement storage capabilities spanning object, block, and file workloads, shared replication, cross-region disaster recovery, and capacity efficiency. Combine hands-on C/C++ development with cross-team delivery, deep systems debugging, and automation of storage operations.

Professional Experience

Oracle Corporation | Jun 2016 - Present

Oracle Cloud Infrastructure | Unified Storage and Block Storage

- Design and drive cross-team initiatives for a unified storage platform serving object, block, and file workloads, including a shared replication layer and erasure coding for object storage.
- Build block-storage support on the unified stack and drive migration from the existing backend; contribute to deep archival capabilities for long-term storage.
- Develop cross-region replication for disaster recovery, including asynchronous replication of volume groups. Ownership spans replication and leader election, snapshots, backup, and cross-region replication.
- Implement application-controlled snapshotting and scale the backup backend with service growth, supporting data protection across the storage lifecycle.
- Increase sellable capacity through cold-data compression, reclaiming 20-30% of storage space. Optimize idle-volume backend capacity through work associated with U.S. Patent 11,412,043.
- Implement reclamation of trimmed user blocks to recover unreferenced storage capacity.
- Scale iSCSI serving through multiprocessing and rearchitect the networking subsystem to reduce hops and increase the number of attached volumes.
- Refactor the storage backend to contain failures within a feature, volume, or device. Automate failed-disk handling and detection and handling of single-host failures.
- Drive automated qualification and deployment through CI/CD, together with single-click region buildout to reduce manual operational work.

Professional Experience Continued

Microsoft Corporation | Sep 2005 - May 2016

Azure Storage | Visual Studio | Windows Systems | Device Platforms

- Designed and implemented an RDMA library for a proof-of-concept Azure Storage backend, connecting systems programming experience with storage transport performance.
- Led Visual Studio Cluster Debugger for HPC/MPI, Parallel Debugger, and GPU Debugger projects spanning CUDA, OpenGL, and DirectX. Built teams of five local and ten remote engineers; contributed to hiring and mentoring.
- Implemented engine test libraries for cluster and GPU debuggers and architected a reusable debugger UI test library. Coordinated with vendors to migrate tests across product cycles.
- Diagnosed Windows operating-system, filesystem, disk, and cluster failures through kernel-dump analysis, reverse engineering, and code review. Worked with development teams and customers to resolve root causes.
- Owned end-to-end HoloLens depth-camera functionality from prototype through public announcement. Developed hardware-validation drivers and tools for Surface Hub and Kinect, and worked on Xbox graphics emulation and frame-capture tooling.

Intel Technology India Private Limited | Mar 2002 - Sep 2005

Systems Software and Hardware Validation

- Designed and developed CPU-core and cache/memory-subsystem validation tools, including Linux kernel programming and drivers for test cards and chipsets.
- Investigated hardware and operating-system defects, drove issues to closure, and wrote targeted patches. Developed network-card management and configuration software for Windows and Linux.

Additional Cloud Service Delivery

At Oracle, designed and developed a near-real-time marketplace software-metering service from prototype to public release. Helped redirect and implement a data-processing service, taking it from prototype through internal release.

Technical Skills

Languages: C/C++; Java, Python, and C# as needed.

Storage and systems: Object, block, and file storage; replication; leader election; erasure coding; snapshots; backup; disaster recovery; compression; iSCSI; RDMA; Linux kernel and device drivers; concurrent and parallel programming.

Infrastructure and tooling: Kubernetes, Terraform, Docker, Helm, Prometheus, gRPC, Folly, Boost, ZooKeeper, Kafka, and Flink.